# TELECOM CHURN ANALYSIS

*Retention Diagnostic & Recommendations*

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# Contents

# Executive Summary

## How to read this report

This dataset shows an overall churn rate of 88.9% — substantially higher than typical telecom benchmarks (15–30%). Because almost every customer in this sample has churned, this report focuses on relative comparisons — which customer segments churn more or less than this elevated baseline — rather than treating any single rate as “low.” Every finding below should be read as “better or worse than a very high norm,” not “good or bad in absolute terms.”

## The headline finding

Retention in this business is not about fixing individual customers — it is about getting more customers into the one combination that already works. Customers on annual or biennial contracts, at low-to-moderate pricing, with Tech Support enrolled, churn 33–46 points below the dataset average. That is the only reliably low-churn profile identified in the analysis, and it is a repeatable, targetable combination — not a coincidence of any one variable.

## What's driving churn

- Contract length is the single strongest lever (23.9-point spread between best and worst) — month-to-month customers churn almost universally regardless of any other factor.
- Tech Support is the second strongest lever (21.7-point spread) — but it only moves the needle for customers already on a longer contract. For month-to-month customers, Tech Support makes no measurable difference.
- Price compounds both effects — even the best-performing segment (long contract + Tech Support) loses much of its retention advantage once price moves into the High tier.

## Recommendations

### Recommendation 1: Bundle Tech Support at the point of annual contract signup — not before

Tech Support only reduces churn for customers already committed to a One-Year or Two-Year contract. Pushing it as a universal upsell wastes effort on month-to-month customers, where it has no effect. Attach it specifically at contract renewal or upgrade conversations.

### Recommendation 2: Treat month-to-month as a conversion problem, not a retention problem

No combination of price or service tested reduces churn for month-to-month customers — they churn at or near 100% regardless. Rather than spending retention budget trying to fix this segment in place, prioritize incentives (discounts, bundled Tech Support, contract-switch promotions) that move them onto annual contracts, where the data shows real retention is achievable.

### Recommendation 3: Protect the best-performing segment from price creep

Long-contract, Tech-Support-enrolled customers are the most resilient segment in the dataset — but High pricing erodes most of that advantage. Targeted discounting for this specific group is likely to preserve more revenue through retention than it costs in margin.

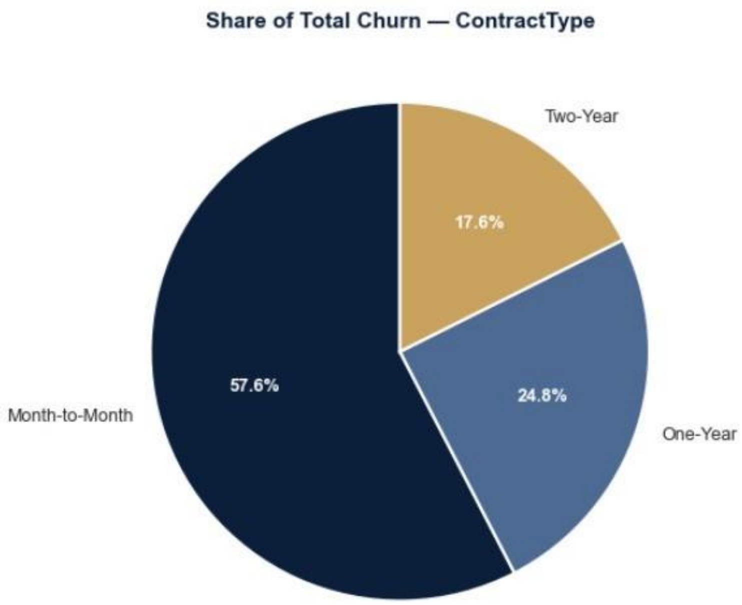
## A. Why Is Churn So High Almost Everywhere?

Before drilling into specific segments, it helps to understand the two factors that set the overall tone of this dataset: contract commitment and price.

### Contract length is the strongest single differentiator

| Contract Type | Churn Rate (%) |
|---|---|
| Month-to-Month | 100.0 |
| One-Year | 76.1 |
| Two-Year | 79.0 |

**So what:** Month-to-month customers churn at a near-universal rate. Annual and biennial contracts, by contrast, sit meaningfully lower — confirming that contract length is the foundation any retention strategy should be built on.

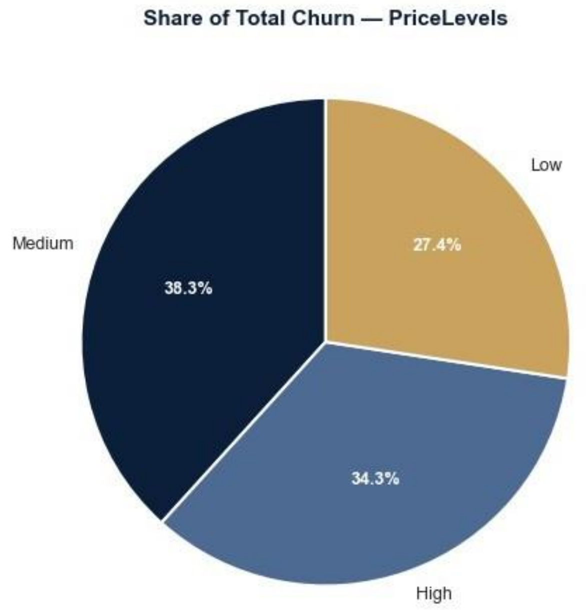



*Share of Total Churn by Contract Type*

### Price adds a secondary, compounding effect

| Price Level | Churn Rate (%) |
|---|---|
| Low ($0–$55) | 84.8 |
| Medium ($55–$90) | 87.5 |
| High ($90–$120) | 95.0 |

**So what:** Every price tier increase adds roughly 5 points of churn. Price alone is a smaller lever than contract length, but it consistently erodes retention on top of whatever contract or service factors are already in play.

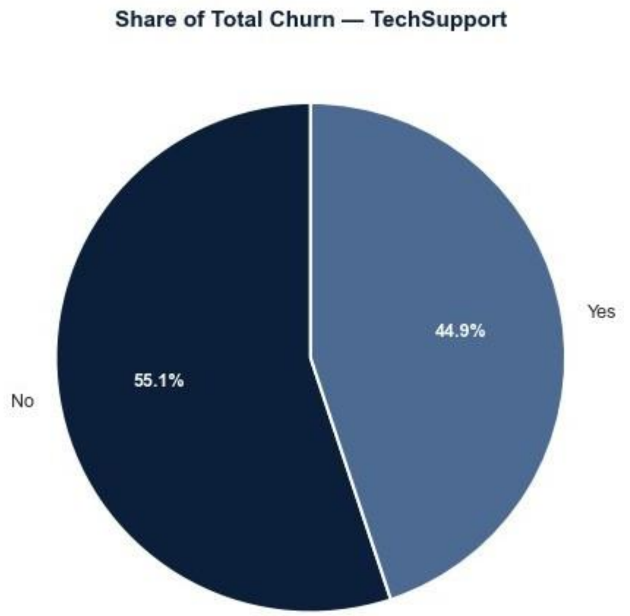



*Share of Total Churn by Price Level*

## B. The Single Biggest Lever: Tech Support — With One Condition

| Tech Support | Churn Rate (%) |
|---|---|
| Yes | 78.3 |
| No | 100.0 |

**So what:** Every customer without Tech Support in this dataset has churned, regardless of contract type, price, or internet service. But Tech Support only becomes a real retention tool when it's paired with a longer contract — see the heatmap in Section D for exactly where this holds.

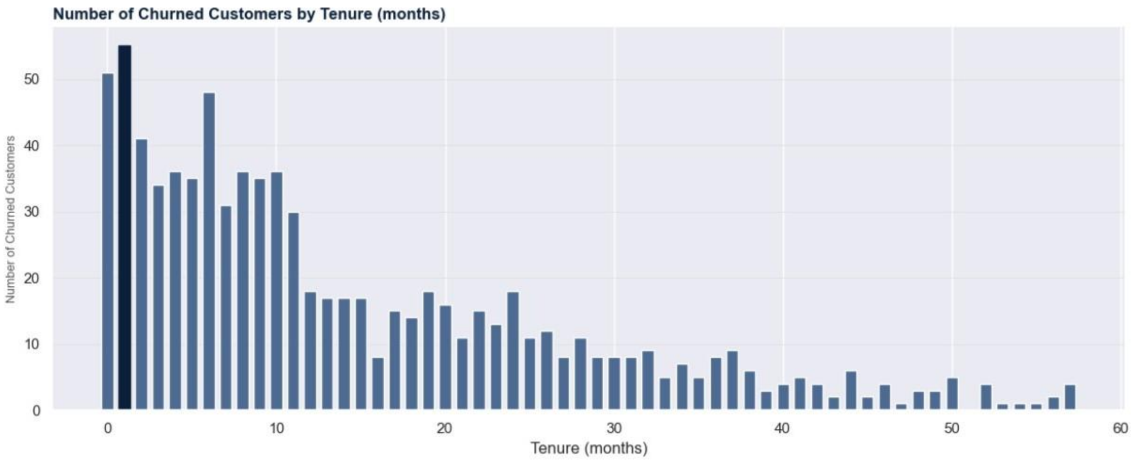



*Share of Total Churn by Tech Support Status*

## C. Who Is Leaving Early, and Does It Matter?

Just over half of all churned customers (51.75%) leave within their first 12 months — and this early churn splits almost evenly between month-to-month customers (27.15%) and customers who were on annual/biennial contracts (24.60%).

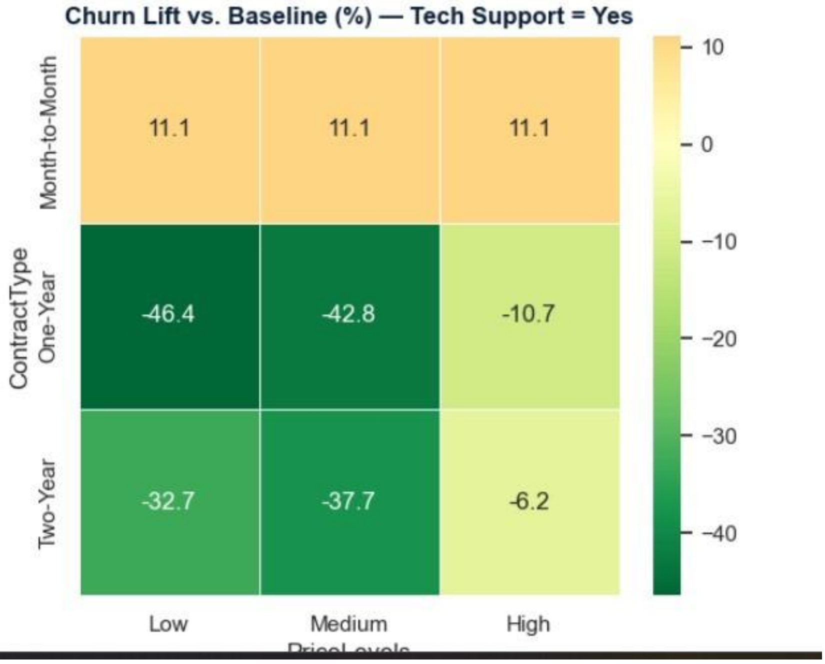
**So what:** Early tenure is a period of real vulnerability across contract types, not just among month-to-month customers. This suggests onboarding and early-relationship engagement (not just contract type) is worth its own attention — a customer signing a Two-Year contract is not automatically “safe” in the first year.




*Number of Churned Customers by Tenure (months)*

## D. Where Should Retention Effort Actually Go?

This is the combined view: which specific combinations of Contract Type and Price Level, when Tech Support is enrolled, perform best against the 88.9% baseline.




*Churn Lik vs. Baseline (%), Tech Support = Yes*

**So what:** One-Year and Two-Year contracts at Low or Medium price, with Tech Support enrolled, churn 33–46 points below baseline — by far the strongest-performing segment identified anywhere in this analysis. This is the profile to actively sell into and retain, not just the one to "not worry about." High price erodes this advantage sharply even within the same contract/support combination.

*For month-to-month customers, none of these levers move the needle — Tech Support, price, and internet type all produced ~100% churn in that segment. See Recommendation 2: this segment needs a conversion strategy, not a retention one.*

# Appendix: Supporting Data & Methodology

The tables and charts below provide the full underlying detail behind the findings summarized in this report, including sample sizes, statistical reliability checks, and data quality notes.

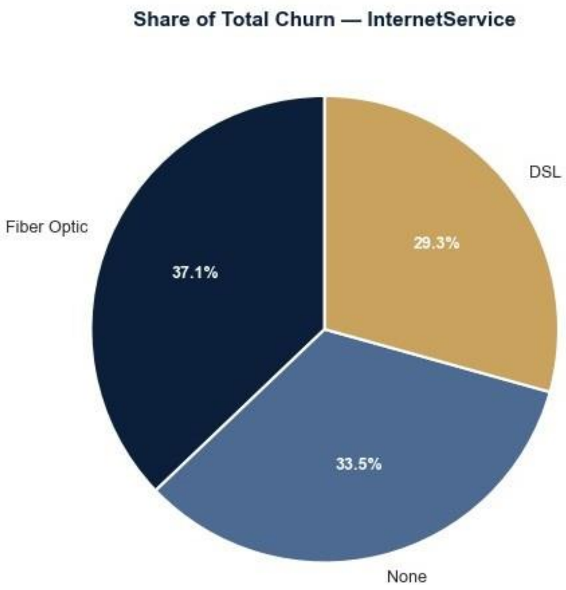
## A1. Monthly charges by customer group

| Customer Group | Avg. Monthly Charges |
| --- | --- |
| All customers | 73.96 |
| Churned | 78.23 |
| Non-churned | 60.78 |


*Distribution of Monthly Charges by Churn Status*

## A2. Internet Service

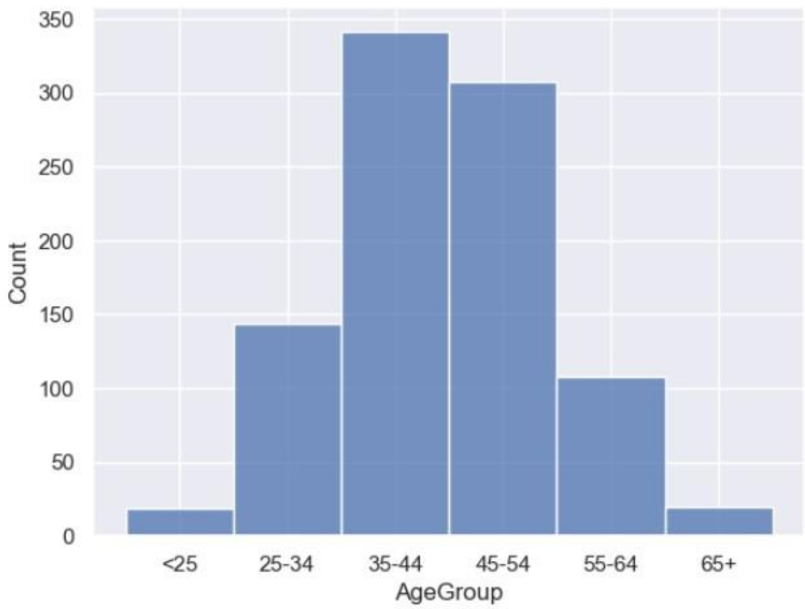
| Internet Service | Churn Rate (%) |
| --- | --- |
| Fiber Optic | 84.5 |
| DSL | 84.0 |
| None | 100.0 |




*Share of Total Churn by Internet Service*

## A3. Customer profile — age, gender, tenure

| Age Bracket | Churn Rate (%) |
|---|---|
| <25 | 78.9 |
| 25–34 | 88.2 |
| 35–44 | 88.9 |
| 45–54 | 88.3 |
| 55–64 | 91.7 |
| 65+ | 100.0 |



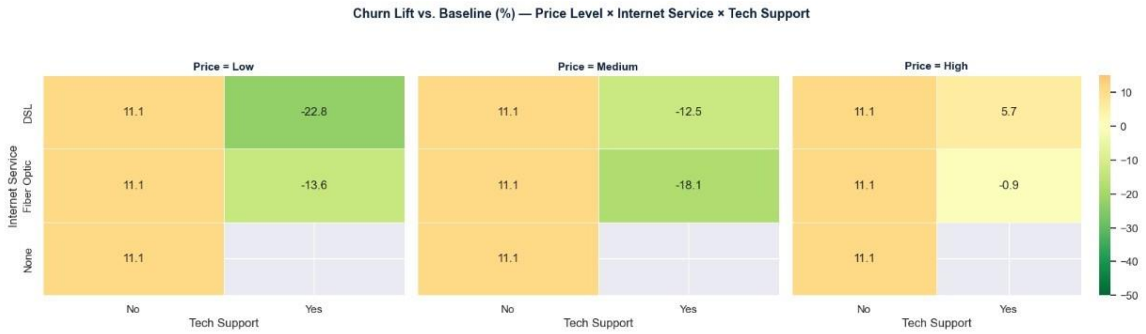

*Customer Count by Age Group*

| Gender | Churn Rate (%) |
|---|---|
| Male | 88.3 |
| Female | 89.7 |
| **Tenure (months)** | **Churn Rate (%)** |
| <10 | 100.0 |
| 10–20 | 79.1 |
| 20–30 | 80.4 |
| 30+ | 78.3 |

**Median tenure (overall):** 13 months

**Median tenure (churned):** 11 months

### A4. Price × Internet Service × Tech Support




*Churn Lik vs. Baseline (%): Price Level × Internet Service × Tech Support*

### A5. Reliability check — sample size and dataset share

Sample size (n) and dataset share for every Contract Type × Price Level × Tech Support combination shown in Section D, confirming each plotted cell is statistically reliable.

| ContractType | PriceLevels | TechSupport | n | churn_count | churn_rate_pct | dataset_share_pct | lift_vs_baseline | reliable |
|---|---|---|---|---|---|---|---|---|
| Month-to-Month | Medium | No | 107 | 107 | 100.0 | 11.4 | 11.1 | True |
| Month-to-Month | Medium | Yes | 88 | 88 | 100.0 | 9.4 | 11.1 | True |
| Month-to-Month | High | Yes | 81 | 81 | 100.0 | 8.6 | 11.1 | True |
| Month-to-Month | High | No | 80 | 80 | 100.0 | 8.5 | 11.1 | True |
| Month-to-Month | Low | Yes | 63 | 63 | 100.0 | 6.7 | 11.1 | True |
| Month-to-Month | Low | No | 62 | 62 | 100.0 | 6.6 | 11.1 | True |
| One-Year | Medium | Yes | 52 | 24 | 46.2 | 5.5 | -42.7 | True |
| One-Year | Medium | No | 51 | 51 | 100.0 | 5.4 | 11.1 | True |
| One-Year | Low | Yes | 47 | 20 | 42.6 | 5.0 | -46.3 | True |
| One-Year | High | Yes | 46 | 36 | 78.3 | 4.9 | -10.6 | True |
| Two-Year | Medium | Yes | 41 | 21 | 51.2 | 4.4 | -37.7 | True |
| One-Year | High | No | 41 | 41 | 100.0 | 4.4 | 11.1 | True |
| One-Year | Low | No | 35 | 35 | 100.0 | 3.7 | 11.1 | True |
| Two-Year | Low | Yes | 32 | 18 | 56.2 | 3.4 | -32.7 | True |
| Two-Year | Low | No | 31 | 31 | 100.0 | 3.3 | 11.1 | True |
| Two-Year | Medium | No | 29 | 29 | 100.0 | 3.1 | 11.1 | True |
| Two-Year | High | Yes | 29 | 24 | 82.8 | 3.1 | -6.1 | True |
| Two-Year | High | No | 24 | 24 | 100.0 | 2.6 | 11.1 | True |

*Sample Size and Dataset Share by Combination*

> *18 of 18 combinations are statistically reliable (n ≥ 15), covering 100.0% of the dataset.*

### A6. Ranked driver table

| Variable | Min Rate | Max Rate | Spread |
|---|---|---|---|
| Contract Type | 76.1 | 100.0 | 23.9 |

| Variable | Min Rate | Max Rate | Spread |
|---|---|---|---|
| Tech Support | 78.3 | 100.0 | 21.7 |
| Internet Service | 84.0 | 100.0 | 16.0 |
| Price Level | 84.8 | 95.0 | 10.2 |
| Gender | 88.3 | 89.7 | 1.5 |

### A7. Lift vs. baseline (88.9%), full detail

| Internet Service | Lift vs. Baseline |
|---|---|
| DSL | −4.42 |
| Fiber Optic | −4.92 |
| None | +11.02 |
| **Contract Type** | **Lift vs. Baseline** |
| Month-to-Month | +11.08 |
| One-Year | −12.82 |
| Two-Year | −9.92 |
| **Price Level** | **Lift vs. Baseline** |
| Low | −4.12 |
| Medium | −1.92 |
| High | +6.07 |

### A8. Top highest-risk, highest-contribution intersections

| Contract | Price | Internet | Tech Supp. | Churn % | n | % of Total Churn |
|---|---|---|---|---|---|---|
| Month-to-Month | Medium | None | No | 100 | 61 | 7.90 |
| Month-to-Month | High | Fiber Optic | Yes | 100 | 51 | 6.11 |
| Month-to-Month | Medium | Fiber Optic | Yes | 100 | 48 | 5.75 |
| Month-to-Month | High | None | No | 100 | 48 | 5.75 |
| Month-to-Month | Medium | DSL | Yes | 100 | 40 | 4.79 |

### A9. Data quality documentation

- TotalCharges was excluded from driver analysis, as it is derivative of Tenure × Monthly Charges rather than an independent variable.
- The dataset is highly skewed, with an 88.9% overall churn rate — substantially above typical industry benchmarks — which is why all segment-level findings are reported as lift vs. baseline rather than raw rate.
- Tech Support = No customers churn at 100% regardless of Contract Type, Price Level, or Internet Service; this segment is reported as a single summary line rather than included in the Section D heatmap grid, which covers Tech Support = Yes only.

- InternetService = None and TechSupport = No are not fully independent — every customer without internet service also lacks tech support by definition — so their individual spreads in A6 should not be read as two additive, independent churn drivers.